

Is Costa Rica's Transition Fast Enough for 1.5°C?

CO₂, GDP and Transport in a Supposed Green Champion

Group PG-20

Plan

1. **Q1 – Are we on a 1.5°C path?**

If we assume a simple linear decline in CO₂ per capita from 2010 to net-zero by 2050, how far is Costa Rica above or below that pathway today?

2. **Q2 – Decoupling compared with similar economies**

Among countries that had similar GDP per capita to Costa Rica in 2010, has Costa Rica done unusually well (or not) at growing income while stabilising or reducing CO₂ per capita?

3. **Q3 – Is transport now the main bottleneck?**

Given that Costa Rica's electricity is already mostly low-carbon, to what extent are transport emissions now dominating its remaining CO₂ and limiting further progress?

Data & approach

- **Country focus:** Costa Rica
- **Core data:** CO₂ emissions, CO₂ per capita, GDP per capita, and sectoral emissions
- **Source:** Our World in Data – CO₂ and Energy
- **Period:** From 1960s to early 2020s, with a focus on 2010 onwards for the 1.5°C pathway

Q1 – 1.5°C pathway

Compare observed CO₂ per capita with a **simple linear net-zero-by-2050 path** starting in 2010.

Q2 & Q3 – Structure and sectors

- Build a **peer group** of countries with similar GDP per capita in 2010
- Compare **GDP vs CO₂ per capita changes** and then zoom in on **transport vs other sectors** for Costa Rica

Q1 – Are we on a 1.5°C path?

- I use a **simple stylised pathway**: CO₂ per capita declines **linearly** from 2010 to **net-zero in 2050**.
- Costa Rica's **observed CO₂ per capita** is then plotted against this pathway.
- The **gap in the latest year** shows how far Costa Rica is **above** the 1.5°C-style path, under these assumptions.
- This is not a full carbon budget model – it's a **transparent, back-of-the-envelope benchmark**.

